# NEET Made Simple

Easy explanations + practice questions (like NEET asks)

This booklet explains important topics in **simple, everyday words** — no heavy textbook language. After each topic, there are practice questions written in the same style as NEET (same difficulty, same question pattern) so you can test yourself.

**Important:** These are practice questions made to match NEET's style and difficulty — they are not copied from any real NEET paper (nobody can legally give you a real leaked paper; those Telegram "leak" sellers are scammers). Practicing questions like these is what actually improves your score.

# BIOLOGY

## 1. How our body gets energy (Respiration)

Our cells break down glucose (sugar) to make energy in the form of ATP. This happens in 3 steps: **Glycolysis** (in the cell fluid, glucose breaks into pyruvate), **Krebs cycle** (inside mitochondria), and **Electron Transport Chain / ETC** (also in mitochondria, makes most of the ATP). Total ATP from one glucose molecule (aerobic) is about 36-38.

**Q1. Where does the Krebs cycle take place in a cell?**

A) Cell membrane
B) Mitochondrial matrix
C) Nucleus
D) Cytoplasm

**Answer: B**

Why: The Krebs cycle happens inside the mitochondria, specifically in the matrix (the fluid inside the inner membrane).

**Q2. Which process gives the maximum number of ATP molecules?**

A) Glycolysis
B) Krebs cycle
C) Electron Transport Chain (ETC)
D) Fermentation

**Answer: C**

Why: ETC produces the most ATP (around 32-34 out of the total 36-38) because it uses the electrons collected earlier to pump out a lot of energy.

## 2. Genetics made simple

Mendel found that traits (like tall/short plant) are controlled by 'factors' we now call genes. Each trait has two copies (alleles) — one from each parent. A **dominant** allele shows its effect even with just one copy; a **recessive** allele only shows when both copies are recessive.

**Q3. In pea plants, tall (T) is dominant over short (t). What will be the phenotype of a Tt plant?**

A) Tall
B) Short
C) Medium height
D) Cannot be determined

**Answer: A**

Why: Since T (tall) is dominant, even one copy of T is enough to make the plant tall, no matter what the other allele (t) is.

**Q4. A colour-blind man marries a woman who is not a carrier. What is the chance their daughter is colour blind?**

A) 0%
B) 25%
C) 50%
D) 100%

**Answer: A**

Why: Colour blindness is X-linked recessive. The daughter gets one normal X from her mother (non-carrier) and one X (with the defect) from her father, making her a carrier but not colour blind herself.

## 3. Hormones in our body (Endocrine system)

Different glands release hormones (chemical messengers) that control body functions. A few you must remember directly: **Insulin** (pancreas) — lowers blood sugar. **Thyroxine** (thyroid) — controls metabolism. **Adrenaline** (adrenal gland) — 'fight or flight' response, increases heart rate.

**Q5. Which hormone is responsible for lowering blood glucose level?**

A) Glucagon
B) Insulin
C) Adrenaline
D) Thyroxine

**Answer: B**

Why: Insulin, released by beta cells of the pancreas, helps cells absorb glucose from the blood, thereby lowering blood sugar level.

**Q6. Deficiency of thyroxine in adults leads to which condition?**

A) Diabetes
B) Goitre / Myxoedema
C) Night blindness
D) Scurvy

**Answer: B**

Why: Low thyroxine (often due to iodine deficiency) can cause goitre (swollen thyroid gland) and myxoedema in adults.

## 4. Plants and sunlight (Photosynthesis)

Plants make their own food using sunlight, water, and $CO_2$, releasing oxygen. It has two parts: **Light reaction** (happens in thylakoid, needs sunlight, makes ATP + NADPH + $O_2$) and **Dark reaction / Calvin cycle** (happens in stroma, doesn't directly need light, uses ATP+NADPH to make sugar).

**Q7. Oxygen released during photosynthesis comes from which molecule?**

A) Carbon dioxide
B) Water
C) Glucose
D) ATP

**Answer: B**

Why: During the light reaction, water molecules are split (photolysis of water), and this releases oxygen as a byproduct.

# CHEMISTRY

## 1. Mole concept — simple explanation

A 'mole' is just a counting unit, like how 'dozen' means 12. One mole of anything = $6.022 \times 10^{23}$ particles (Avogadro's number). To find moles: **moles = given mass ÷ molar mass**.

**Q8. How many moles are there in 44 g of CO2? (Molar mass of CO2 = 44 g/mol)**

A) 0.5 mole
B) 1 mole
C) 2 moles
D) 44 moles

**Answer: B**

Why: Moles = given mass / molar mass = 44/44 = 1 mole.

## 2. Acids, bases and pH

pH tells us how acidic or basic a solution is, on a scale of 0-14. Below 7 = acidic, above 7 = basic, exactly 7 = neutral. Formula: $pH = -\log[H^+]$.

**Q9. A solution has pH = 3. What type of solution is it?**

A) Strongly basic
B) Neutral
C) Weakly basic
D) Acidic

**Answer: D**

Why: Any pH value below 7 means the solution is acidic; the lower the number, the stronger the acid.

## 3. Carbocation stability (Organic chemistry basics)

A carbocation is a carbon atom with a positive charge. The more carbon groups attached to it, the more stable it is, because those groups 'push' electron density towards the positive charge. Order: **tertiary > secondary > primary** carbocation stability.

**Q10. Which carbocation is the most stable?**

A) Primary (1°)
B) Secondary (2°)
C) Tertiary (3°)
D) All are equally stable

**Answer: C**

Why: Tertiary carbocations are most stable because they have three alkyl groups donating electron density and reducing the positive charge's instability.

## 4. Periodic table trends

As you move left to right in a period, atoms get smaller (more protons pull electrons in tighter) and it becomes harder to remove an electron (ionization energy increases). Going down a group, atoms get bigger because more electron shells are added.

**Q11. Which element has the smallest atomic radius: Na, Mg, Al, or Si (all in period 3)?**

A) Na
B) Mg
C) Al
D) Si

**Answer: D**

Why: Moving left to right across a period, atomic radius decreases because the increasing nuclear charge pulls electrons closer. Si is furthest right, so it's the smallest here.

# PHYSICS

## 1. Newton's Laws — in plain words

1st law: an object keeps doing what it's doing (resting or moving) unless a force acts on it. 2nd law: Force = mass × acceleration ($F = ma$) — bigger force means bigger acceleration. 3rd law: every action has an equal and opposite reaction.

**Q12. A force of 10 N acts on a 2 kg object. What is its acceleration?**

A) 2 $m/s^2$
B) 5 $m/s^2$
C) 10 $m/s^2$
D) 20 $m/s^2$
**Answer: B**
Why: Using $F = ma$, $a = F/m = 10/2 = 5\ m/s^2$.

## 2. Light bending through lenses

A convex lens bends (converges) light rays to meet at a point; a concave lens spreads (diverges) them. Lens formula: $1/v - 1/u = 1/f$, where u = object distance, v = image distance, f = focal length.

**Q13. Which type of lens is used to correct short-sightedness (myopia)?**

A) Convex lens
B) Concave lens
C) Cylindrical lens
D) Bifocal lens only
**Answer: B**
Why: In myopia, the eye focuses images before the retina. A concave (diverging) lens spreads the light out slightly so it focuses correctly on the retina.

## 3. Photoelectric effect (light knocking out electrons)

When light of enough energy falls on a metal, it can knock electrons out of it — this is the photoelectric effect. Only light above a certain minimum frequency (threshold frequency) can do this, no matter how bright/dim the light is.

**Q14. In the photoelectric effect, increasing the intensity (brightness) of light (frequency kept same, above threshold) increases:**

A) The energy of each electron
B) The number of electrons emitted
C) The threshold frequency
D) Nothing changes
**Answer: B**
Why: Brightness (intensity) means more photons hitting the metal, so more electrons get knocked out. But the energy of EACH electron depends only on the frequency of light, not brightness.

## 4. Current electricity basics

Ohm's law: $V = IR$ (Voltage = Current × Resistance). Resistors in **series** add up directly ($R = R1+R2+...$). Resistors in **parallel** add up as reciprocals ($1/R = 1/R1 + 1/R2 + ...$).

**Q15. Two resistors of 4 Ω each are connected in parallel. What is the total resistance?**

A) 8 Ω
B) 4 Ω
C) 2 Ω

D) 1 Ω

**Answer: C**

Why: For parallel: $1/R = 1/4 + 1/4 = 1/2$, so $R = 2\ \Omega$.

## Before You Go

- Read the simple explanation first, then try the question yourself before checking the answer.
- If you get it wrong, re-read the explanation slowly — don't just memorise the answer letter.
- Do this daily with small topics rather than cramming everything in one day.
- Stay away from anyone online claiming to sell a 'leaked' NEET paper — it's always a scam, and NTA has confirmed this repeatedly.

All the best — steady daily practice beats last-minute panic!